A Crowdsourced Civic Issue Reporting and Resolution System with AI-Based Prioritization

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***Abstract*—There are many civic issues that cities encounter like potholes, waste buildup, and infrastructure failure. Currently available complaint systems have faced challenges of inefficiency, opacity, and low participation rate of citizens. This paper describes the development of a crowdsourced civic issue reporting and resolving system that combines the usage of mobile technology and web platforms together with a priority algorithm powered by AI.**

With this proposed system, citizens can report problems via mobile application using images, geolocation data, and textual description. Reports can be categorized based on priority by applying the algorithm which uses multiple criteria such as text, location and timing factors. Moreover, gamification technique can be used to encourage users in submitting useful reports.

Experimental analysis has shown the improvement in effectiveness of handling civic issues using this system. The approach can be successfully applied in smart cities environment.

Index Terms—Civic Issues, Smart City, AI Prioritization, Crowdsourcing, Gamification, Geospatial Analysis

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I. Introduction

The rise in urbanization has resulted in many civic problems like damaged roads, improper management of wastes, water leaks, and electrical faults. Conventional citizen complaint methods such as helpline numbers and websites have proven to be inefficient as they suffer from various inefficiencies such as delays, lack of transparency, and lack of citizen involvement.

In this era where mobile phones are widely used by people, many online platforms are developed to report civic problems. Nonetheless, most of the conventional civic complaint systems lack an effective way of prioritizing and solving these complaints.

In this paper, a civic issue reporting and solving system is proposed. The system uses smart phones, cloud computing, and artificial intelligence techniques in prioritizing and solving civic issues. This paper makes the following contributions: 1. Multi-factor prioritization framework that considers text, location, and time, 2. Gamification techniques to increase public involvement, 3. Cloud infrastructure that allows for real-time processing of the data, and 4. Transparency by tracking workflow processes and their status.

II. Literature Survey

Civic issue reporting systems have gained considerable importance in smart cities since they contribute to enhancing communication between government officials and citizens. Traditional applications, like web portals or hotlines, enable citizens to register complaints, however, lack transparency, efficiency of processing and solving reported issues.

The rise of new technologies such as GIS and mobile application development have allowed implementing solutions enabling users to send their notifications based on images, descriptions and geographical coordinates of problem areas. Although current systems benefit from applying GIS analysis, they mostly employ human labor, resulting in ineffective processing of high loads of data.

The use of cloud computing technologies has allowed modernizing civic management applications. They are beneficial due to scalability and instant access to required data provided by backend services like authentication, database management and storage. Despite the advantages provided by cloud technologies, most of the existing systems do not make use of cloud computing capabilities in terms of decision-making and data automation.

Some researchers have also implemented AI-based methods like NLP for analyzing and categorizing user submissions. Such methods prove to be helpful in improving the processing of data. Still, the use of multi-factor prioritization based on text analysis, geographical location and time has not been widely discussed yet.

As far as user engagement is concerned, studies have shown that participation tends to decline over time. Therefore, some systems employ gamification techniques including reward points and ranking.

In conclusion, although current civic issue reporting solutions possess certain functionality, there are not enough integrated tools providing intelligent data classification, automation processes and user engagement through gamification.

III. Methodology

The proposed system follows a structured approach for reporting, analyzing, and resolving civic issues using mobile and web technologies supported by a cloud-based backend.

The methodology ensures efficient data collection, processing, and communication among citizens, administrators, and government departments.

A. System Architecture

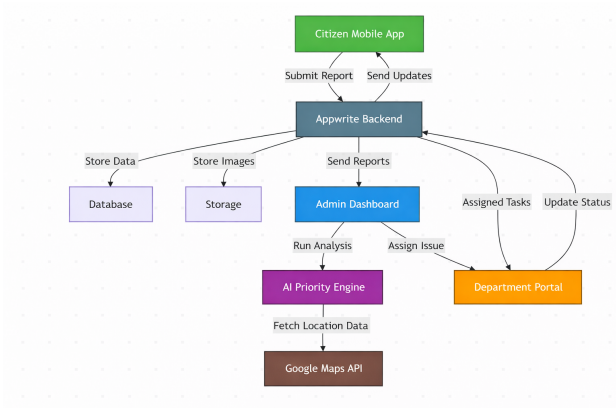

Fig. 1. System Architecture of the Proposed Civic Issue Reporting System

As can be seen from Fig. 1, the system comprises of three main elements which include the mobile application, the administrative dashboard, and the departmental portal. These elements are integrated using a cloud-based backend service that ensures the efficiency of data exchange and real-time communication.

As indicated by the architectural design of the system, it includes some important elements that work together to provide effective operation of the entire system. From Fig. 1 below, it can be seen that the system comprises of three major parts namely; a mobile application used to report problems, an administrative dashboard to monitor and validate reports, and a departmental portal responsible for resolving problems.

B. Report Submission

Citizens are free to report civic problems using the mobile app using an image and a brief description, together with the location of the problem. Location information is acquired via GPS services, while images will be stored in cloud storage. All this data is kept in one centralized database.

This strategy ensures that all data collected is accurate.

C. AI-Based Prioritization

In order to manage a large amount of reports effectively, the system uses an intelligent algorithm for prioritizing based on the following three important aspects: description, location, and peak hours of issue reporting.

$$Priority = \alpha(D) + \beta(L) + \gamma(P) \tag{1}$$

Here,

- D is the degree of importance inferred from the description provided by users (”accident,” ”danger,” etc.),
- L is the location aspect with higher priority for problems occurring near sensitive locations (hospitals, schools, busy streets), and
- P is the peak time aspect, when incidents are assigned greater priority if they occur during busy or peak hours.

According to the calculated priority, reports are divided into high, medium, and low priority reports.

D. Workflow Management

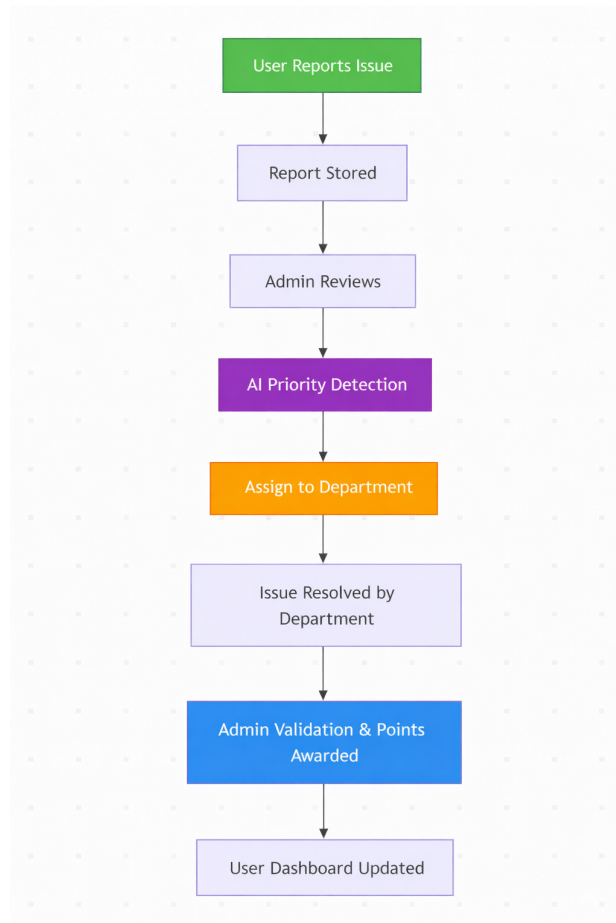

Fig. 2. Workflow of Civic Issue Reporting and Resolution

A diagram representing the workflow of the system is shown in Fig. 2. After issue submission by the user, validation, prioritization, and assignment to the concerned department take place. The status is tracked till the problem is sorted out.

The workflow of the system helps in efficient handling of any issue. Fig. 2 depicts the workflow in brief.

Upon issue submission, it is first validated by the administrator to check for its validity. This is followed by prioritizing and assigning the issue to the concerned department. It is handled by the concerned department with status updates at various stages of completion.

Here is the workflow of the entire process.

E. Gamification Mechanism

The gamification method is introduced in order to engage users in the process. The number of points for each user depends on the precision and helpfulness of their reports, and

rewards are received instantly once validated, which makes participation more attractive.

Furthermore, the penalty method is used in order to ensure that false reports do not occur frequently.

F. Technology Stack

This makes use of sophisticated technology, which ensures efficiency and scalability. The mobile application is developed using Flutter framework, while the admin and department portal uses React. The backend uses BaaS (backend as a service), which runs on the cloud and helps in carrying out database, authentication, and storage operations.

Also, mapping APIs have been used to enable geolocation capabilities and spatial analysis, which allows us to determine the exact location of the problem area.

IV. Results and Discussion

The implementation of the suggested framework was conducted using mobile and web technologies, and its assessment was done in simulated real-life situations. This evaluation will consider system performance, efficiency, and usability.

A. System Performance

This system can efficiently allow users to register complaints about civic issues via image and description uploads and geolocation data using the mobile app. The server-side database guarantees the proper storage of information, and it supports the effective interaction between the mobile app, the admin interface, and the department site.

The prioritization function processes reports according to their user description, location, and peak time status. It allows for issue classification into several priority categories.

B. Workflow Efficiency

TThe workflow followed by him ensures that the problems are processed seamlessly from their submission to resolution. Upon submission of a report, the administrator validates and assigns it to the relevant department. The status of the report is updated at various levels.

The application of an automatic prioritization method decreases the workload associated with manually organizing the reported problems.

C. User Engagement

The gamification model motivates the users to engage in submitting civic problems. They earn rewards for the content and relevancy of the report, which encourages them to make effective submissions.

The introduction of a penalty model decreases erroneous submissions by discouraging users from submitting misleading problems.

D. Discussion

The outcome proves that the designed framework acts as a reliable channel for reporting civic issues. The combination of cloud computing and the prioritization technique helps coordinate communication between users and officials.

Nevertheless, the existing framework faces some drawbacks. The process of prioritization is mainly rule-oriented and might overlook several factors in reality. Moreover, the framework lacks live notifications, which should be added to the system in future iterations.

In summary, the developed framework has promising implications in the smart city domain.

V. Conclusion and Future Scope

This paper explores the design and implementation of a citizen civic issue reporting and resolving platform that uses the power of mobile technology, cloud computing, and an artificial intelligence inspired prioritization method.

In this paper, a crowd-sourcing model is explored in which citizens report the civic issues using photographs and a text description of the problem while administrators and concerned departments handle their reporting based on a predefined workflow.

An innovative approach to prioritize civic issues has been proposed here that is dependent on various parameters including the user description, issue location, and peak time conditions among others. The implementation of a gamification module has ensured the user cooperation and provided the motivation for users to submit more valuable reports on civic issues.

This civic issues reporting system proves to be highly useful and efficient in managing various civic issues. In addition, its use of cloud computing infrastructure makes it scalable and suitable to use in the context of smart cities.

Although this system has achieved the goal in question, yet there are some aspects that may need further refinement. The prioritization method implemented here is rule-based and lacks adaptability as it cannot deal with many of the complex situations. Another drawback is that it doesn't support any real-time notifications feature.

Machine learning models can be included in the priority method to improve its accuracy and adaptability. Image-based validations can help confirm the credibility of submitted civic issues by users. Other improvements may include implementing real-time notification and visualization features such as heatmaps.

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